

Formal Elements of Game Design

This Week

- Mechanics Dynamics Aesthetics – structure of games
- Formal Elements of Game Design

Mechanics, Dynamics, Aesthetics

Would you like to know more?

Read:

Robin Hunicke, Marc LeBlanc, Robert Zubek - MDA: A Formal Approach to Game Design and Game Research



Activity

Discussion: MDA of Chess

In Chess, what are examples of:

1. Mechanics
2. Dynamics
3. Aesthetics

Mechanics, Dynamics and Aesthetics – The DNA of a Game (continued)

Mechanics

- Mechanics refer to the **rules**, **systems**, and **interactions** that govern how the game functions. They encompass the tangible elements of a game, such as the **actions players can perform**, the goals they need to achieve, and the resources or tools at their disposal.

- **Mechanics** establish the framework within which players can interact with the game world and other players.
  - movement controls
  - combat systems
  - puzzles
  - item management
  - resource allocation

- Effective mechanics provide clear and meaningful choices to players, balancing challenge and accessibility.

Mechanics, Dynamics and Aesthetics – The DNA of a Game (continued)

Dynamics

- Dynamics are the **emergent behaviours** and **interactions** that arise from the mechanics as players engage with the game.
- They are the result of players' actions and decisions **within the defined rule set**.
- Dynamics are often unique to each playthrough, as they depend on the individual choices made by players.
- These can include **strategies**, **tactics**, **social interactions**, and the overall flow and pacing of the game.
- Dynamics can create a sense of immersion, challenge, and depth, as players discover and exploit the possibilities within the mechanics.

Mechanics, Dynamics and Aesthetics – The DNA of a Game (continued)

- Aesthetics in game design refers to the **emotional** and **sensory** experiences evoked in players.
- Aesthetics includes the **mood**, **tone**, and **style** of the game, impacting players' emotional responses and engagement.
- **Aesthetics** are not the assets or art style. **Aesthetics** by this definition are the *emotions created by the game*. Of course, assets affect these emotions.
- It also includes the player's roles, or play fantasy, or all the other bits of fun.
- AESTHETIC = FUN

Mechanics, Dynamics and Aesthetics – The DNA of a Game (continued)

The MDA

- It's important to note that **Mechanics**, **Dynamics** and **Aesthetics** are interconnected and influence each other.
- Well-designed **Mechanics** can lead to engaging **Dynamics**, which create an **Aesthetic player experience**
- Successful game design involves finding the **right balance** and alignment between **Mechanics**, **Dynamics** and **Aesthetics** to create a compelling and memorable game.

How to use MDA

1. Try defining the MDA quickly
2. Pick Aesthetic Experience: How you want players to feel
    a. “Funny, Competitive game about Grocery Store Competition”
    b. “Sandbox post-apocalypse survival shooter where every play session surprises me”
3. Develop Mechanics that lead to Dynamics that create the target emotions/aesthetic experience
    a. Maps reset each 3-5 days, which provides new opportunities for the game to develop in different ways, depending where and how players build, keeps one group from dominating for too long

Activity

Answers: Example MDAs of Chess

1. **Mechanics of Chess:**
   a. “The pawn moves one space forward and takes pieces one space diagonally.”
   b. “The Knight moves two along one axis, plus one along another, and can hop over pieces.”
2. **Dynamics of Chess**
   a. “Controlling the middle of the board is important because it features many angles of attack”
   b. “Moves which put the opponent in check force them into reactionary defenses, stifling their ability to develop their board”
   c. “I can fork his Bishop and Knight if I move here”
3. **Aesthetics of Chess**
   a. “The brave Black Knight charges into battle, leaping over the frontline lances and disrupting White’s backline!”
   b. “I am a genius strategist, thinking ten steps ahead of my opponent”
   c. “Holy cow! I was cocky. My opponent was thinking TWENTY turns ahead of ME! GG; WP. Rematch?”

Any others to add?

Activity

Discussion: MDA of Featured Games

What are examples of **Mechanics**,

Dynamics, and

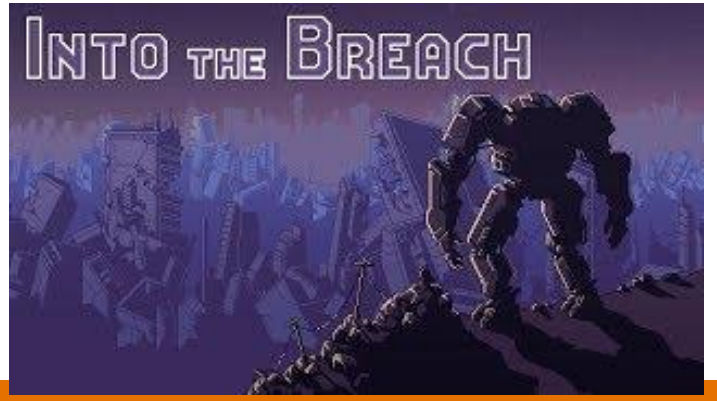
Aesthetics

In these games?

Papers, Please



Into The Breach



Formal Elements of Game Design, as in *Game Design Workshop* by Tracy Fullerton, 2008

Formal Elements of Game Design

- What makes up a game?

There are some common threads…

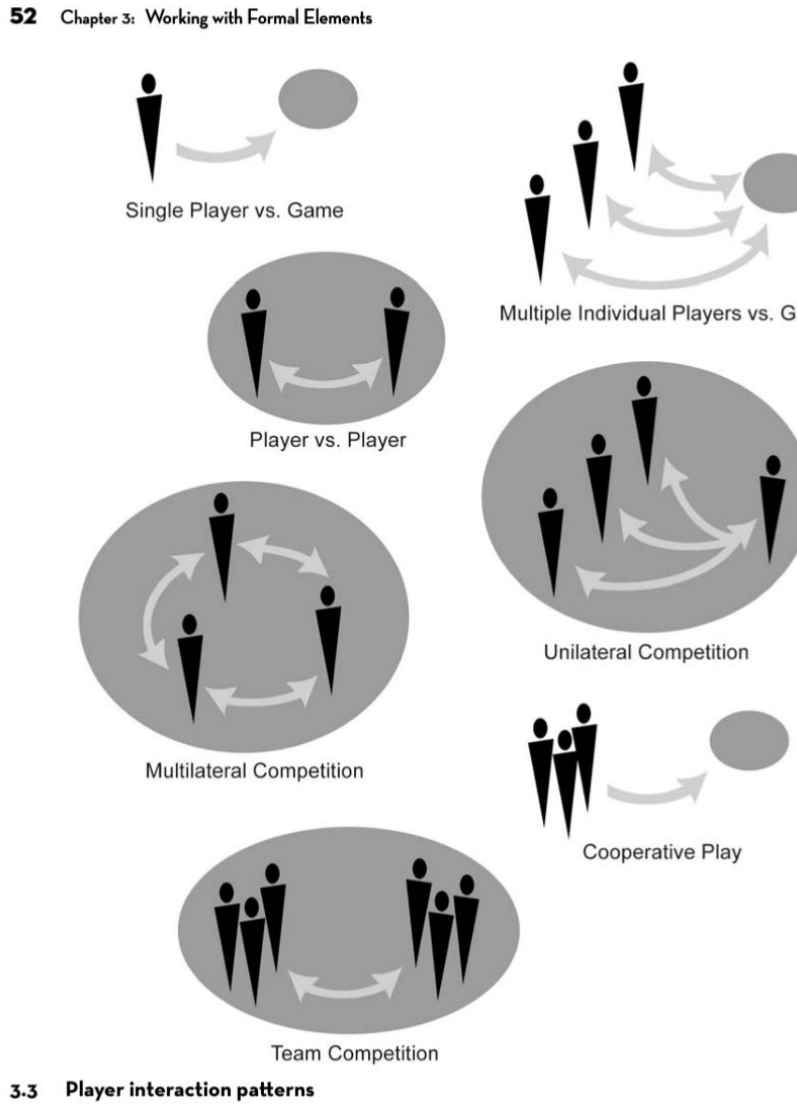
The Formal Elements of Game Design connects these common threads into its definition of required elements of any game. There are 8 parts:

1. Players
2. Objectives
3. Procedures
4. Rules
5. Resources
6. Conflict
7. Boundaries
8. Outcome

Players

Yes. Games have players.

1. How do players enter the game?
2. How many?
3. What roles do they play in the game?
4. Are they all the same role (e.g. SP Story-driven JRPG)?
5. Different roles (e.g. character classes in WOW)?
6. Do roles change?
    a. In Halo's *Infection* gamemode, dying causes you to the zombie team
    b. In Octopus, getting caught by the Octopus turns you into Seaweed
7. How do players interact?
8. Are there computer players? What are they like?



Objectives

What is the GOAL of the game?

1. Say it in ten words or less.
    a. DO NOT say “have fun”. That’s cheating.
2. “The Win condition is…”
3. Is it a positive, negative, or zero-sum game?
    a. Positive-sum: more winners than losers
        i. E.g. DnD, any RPG
    b. Zero-sum: equal winners and losers
        i. E.g. Chess, League of Legends
    c. Negative-sum: more losers than winners
        i. E.g Fortnite, all battle royales
        ii. …or Nuclear War

Procedures

What do you DO in the game?

1. How do you actually proceed?
2. What are the instructions?
3. What are common actions e.g.
    a. Shuffling a deck of cards
    b. Reloading weapons
    c. Saving a game at a Save Point
4. Video games also simulate procedures heavily as functions

Rules

What are the LAWS of the game?

1. **When** are you allowed or not allowed to do → **what** procedures?
   a. **When you pass GO** in Monopoly → **collect $20 from the bank**
2. Rewards for success?
3. Punishments for failure or cheating?
4. Arbitration for ambiguous situations?
5. Systems e.g. Damage over Time effects like Burn, Poison?
6. Restrictions e.g.
   a. PvE/PvP Zones
   b. Level requirements
   c. Technology/Skill Tree requirements

Resources

What do you HAVE in the game?

1. XP, Gold, Items, Units, Abilities…
2. Discrete or Continuous?
3. Tradeable?
4. Sources and Sinks?
5. Economy?
6. Abstract or Concrete Resources?
    a. Abstract Resource: “good/bad team vision”
    b. Concrete Resource: hit points.

Conflict

What STOPS YOU from getting everything you want?

1. What kind of conflict?
2. Do players' goals oppose each other?
3. Do players' goals oppose the computer?
4. Obstacles?
5. Opponents?
6. Tough Decisions?
7. Narrative? Character flaws?

Boundaries

What is PART of the game and what is NOT?

1. Where does the game begin and end?
2. Boundaries of the game world?
3. What are the limits to what you can do?
4. With external websites or location-based games, boundaries of play can exist outside the game world!

Outcome

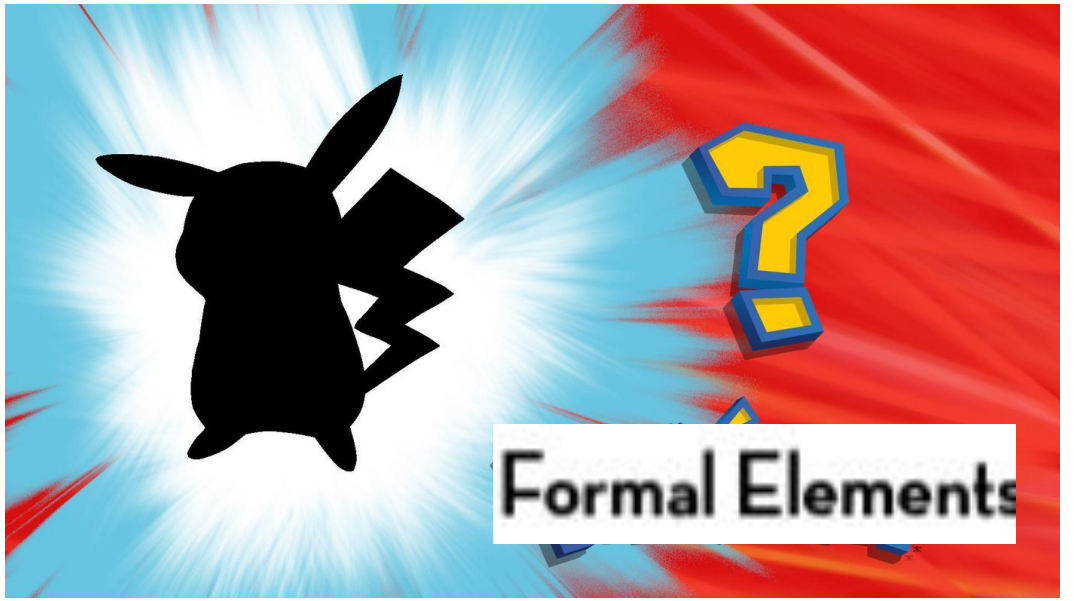
What do you GET OUT in the end?

1. Are there winners and losers?
2. Do you experience a conclusion to the story?
3. Do you produce anything?
    a. Broken Telephone produces an amusing phrase transformation. You can enjoy listening to what it began as, what ended as, and how you really screwed it up

Activity

Quiz: Who's That Formal Element of Game Design?

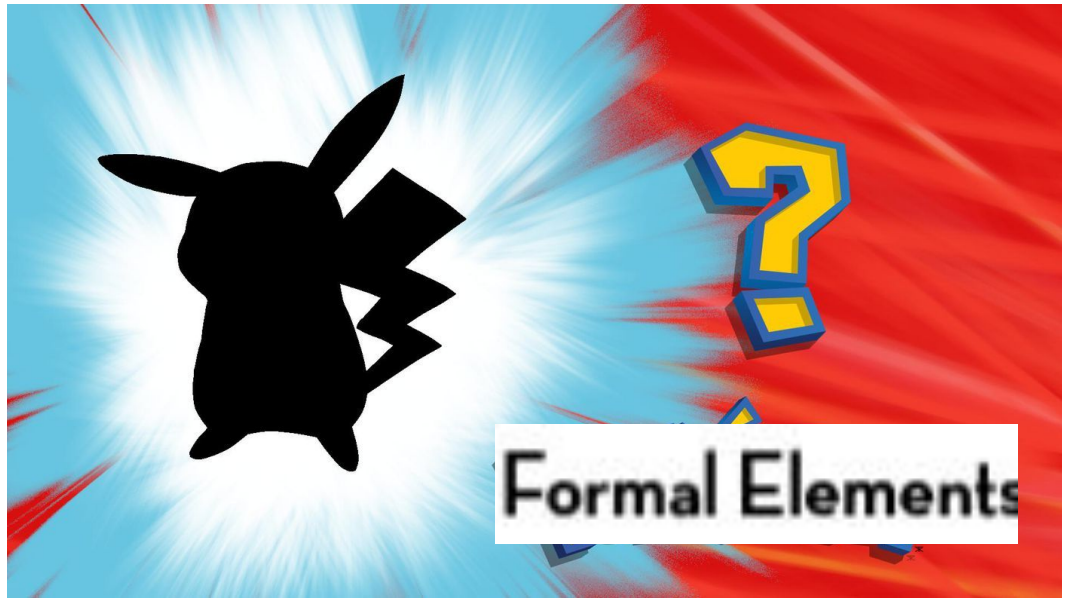
1. When your Pokemon's health drops to zero, it faints
2. A fun story – a memorable moment of genius from your party in Dungeons and Dragons
3. Press Z to Do a Barrel Roll!
4. Good Positioning



Activity

Quiz: Who's That Formal Element of Game Design?

1. When your Pokemon's health drops to zero, it faints
   a. **Rule**
2. A fun story – a memorable moment of genius from your party in Dungeons and Dragons
   a. **Outcome**
3. Press Z to Do a Barrel Roll!
   a. **Procedure**
4. Good Positioning
   a. Abstract **Resource**



Would you like to know more?

Formal & Dramatic Elements of Game Design

- Game Design Workshop by Tracy Fullerton is a major source of the distilled content for this week





Engaging the Player

- **Formal elements** provide structure to the experience of games, but ***what gives these elements meaning for the player?***
- What connects players emotionally to a game experience?
- Games are **entertainment**, and entertainment engages us and moves us both intellectually and emotionally.
- Players become engaged from different things and not all games challenge players equally.
- The **Dramatic Elements of Game Design** are the counterpart that describe why we should care

Wrap-up

The Sum of The Parts

- What may not be apparent from all these elements is the depth to which each element relies on each other.
- Games are **systems**. Systems are groups interrelated elements that work together that form a complex whole. (Whole is greater than the sum of its parts)
  - This means as elements start working together the result is a new dimension of the elements set in motion.

Defining Games

- After outlining **formal** and **dramatic** elements as well as how they combine in **systems** how would we define games?
- Based on these definition we can say about games:
  - A closed, formal system
  - Engages players in a structured conflict
  - Resolves its uncertainty in an unequal outcome.

Beyond Definitions

- Even though we have a basis for what a game is, it is always possible to push the boundaries of those definitions.
- One such area that doesn't totally fit our definition is **serious games**. These are games that **doesn't focus on entertainment but focuses on learning**.
  - Others have called this "gamification" which refers to applying game-like mechanics to more serious simulations.
- There is on-going research that says providing game-like elements to tasks encourage the user to continue and motivates the user

Beyond Definitions (continued)

- Major problem with **gamification** is the “carrot on a stick” conundrum. If you remove the carrot, is there any **motivation** to continue at all?
- Another area of games that moves beyond our definition is **art games**. Some “art games” have hit mainstream success such as *Dear Esther, Firewatch*.
- These are games that don’t necessarily contain conflict but rather focus on the player having an emotional experience.
- Some people do not consider **art games** to be **games**. Some people are dumb. Those are two statements.

Summary

- MDA is a great way to quickly define a game holistically
- Formal Elements make up a game's bits

Would you like to know more?

Try this

In Pairs, select TWO games in the same genre to compare/contrast

- Watch the other play your choice of game
- Consider the MDA of each game. How do they deliver their intended experience to the player?
- I recommend picking games you are **not familiar with** already.

